CODE:

```
import math

# Sample dataset: [Outlook, Temperature, Humidity, Wind], PlayTennis
data = [
    ["Sunny", "Hot", "High", "Weak", "No"],
    ["Sunny", "Hot", "High", "Strong", "No"],
    ["Overcast", "Hot", "High", "Weak", "Yes"],
    ["Rain", "Mild", "High", "Weak", "Yes"],
    ["Rain", "Cool", "Normal", "Weak", "Yes"],
    ["Rain", "Cool", "Normal", "Strong", "No"],
    ["Overcast", "Cool", "Normal", "Strong", "Yes"],
    ["Sunny", "Mild", "High", "Weak", "No"],
    ["Sunny", "Cool", "Normal", "Weak", "Yes"],
    ["Rain", "Mild", "Normal", "Weak", "Yes"],
    ["Sunny", "Mild", "Normal", "Strong", "Yes"],
    ["Overcast", "Mild", "High", "Strong", "Yes"],
    ["Overcast", "Hot", "Normal", "Weak", "Yes"],
    ["Rain", "Mild", "High", "Strong", "No"]
]

features = ["Outlook", "Temperature", "Humidity", "Wind"]

def entropy(rows):
    total = len(rows)
    counts = {}
    for row in rows:
        label = row[-1]
```

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        counts[label] = counts.get(label, 0) + 1
    return sum((-c/total) * math.log2(c/total) for c in counts.values())


def info_gain(rows, col):
    total_entropy = entropy(rows)
    values = set(row[col] for row in rows)
    subsets = [[r for r in rows if r[col]==v] for v in values]
    weighted_entropy = sum((len(s)/len(rows))*entropy(s) for s in subsets)
    return total_entropy - weighted_entropy


def best_feature(rows):
    gains = [info_gain(rows, i) for i in range(len(features))]
    return max(range(len(features)), key=lambda i: gains[i])
def build_tree(rows, depth=0):
    labels = set(r[-1] for r in rows)
    if len(labels)==1: return list(labels)[0]
    if not rows or depth==len(features): return "Unknown"
    col = best_feature(rows)
    tree = {features[col]: {}}
    for v in set(r[col] for r in rows):
        subset = [r for r in rows if r[col]==v]
        tree[features[col]][v] = build_tree(subset, depth+1)
    return tree
tree = build_tree(data)
print("Decision Tree:", tree)
```

OUTPUT:

```
========== RESTART: C:/Users/arunp/OneDrive/Documents/AI/EXP 15.py ==========
Decision Tree: {'Outlook': {'Sunny': {'Humidity': {'High': 'No', 'Normal': 'Yes'}}, 'Rain': {'Wind': {'Weak': 'Yes', 'Strong': 'No'}}, 'Overcast': 'Yes'}}
```